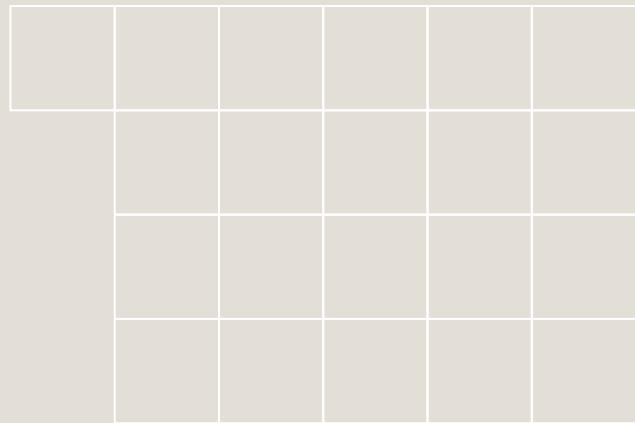


Expense Tracker

(The don't go broke machine)



Necessary Imports

```
import tkinter as tk
```

```
from tkinter import ttk, messagebox
```

```
from datetime import datetime
```

```
FILE_NAME = "expenses.txt"
```

File Handling

```
def add_transaction():  
    t_type = type_var.get()  
    category = category_var.get()  
    amount = amount_entry.get()  
    note = note_entry.get()  
    date = datetime.now().strftime("%Y-%m-%d")
```

File Handling

```
if not amount or not category:
```

```
    messagebox.showerror("Error", "Please fill required fields")
```

```
    return
```

File Handling

```
try:
```

```
    amount = float(amount)
```

```
except ValueError:
```

```
    messagebox.showerror("Error", "Invalid amount")
```

```
    return
```

File Handling

```
with open(FILE_NAME, "a") as f:
```

```
    f.write(f"{date} | {t_type} | {category} | {amount} | {note}\n")
```

```
messagebox.showinfo("Success", "Transaction added")
```

```
load_data()
```

File Handling

```
def load_data():  
    for row in tree.get_children():  
        tree.delete(row)  
  
    try:  
        with open(FILE_NAME, "r") as f:  
            lines = f.readlines()  
    except FileNotFoundError:  
        return
```

File Handling

for line in lines:

```
tree.insert("", "end", values=line.strip().split(" | "))
```

```
update_balance()
```

```
def delete_transaction():
```

```
    selected = tree.selection()
```

```
    if not selected:
```

```
        return
```


File Handling

```
values_to_delete = tree.item(selected)["values"]
```

```
    tree.delete(selected)
```

```
# Rewrite file without deleted row
```

```
with open(FILE_NAME, "r") as f:
```

```
    lines = f.readlines()
```

File Handling

```
with open(FILE_NAME, "w") as f:
```

```
    for line in lines:
```

```
        if line.strip() != " | ".join(map(str, values_to_delete)):
```

```
            f.write(line)
```

```
update_balance()
```

File Handling

```
def update_balance():  
    income = 0  
    expense = 0  
  
    for row in tree.get_children():  
        values = tree.item(row)["values"]  
        t_type = values[1]  
        amount = float(values[3])
```

File Handling

```
if t_type == "Income":
```

```
    income += amount
```

```
else:
```

```
    expense += amount
```

```
balance = income - expense
```

```
balance_label.config(text=f"Balance: {balance:.2f}")
```

User Interface(UI)

```
root = tk.Tk()
```

```
root.title("Expense Tracker (Text File)")
```

```
root.geometry("750x500")
```

```
# Input Frame
```

```
frame = tk.Frame(root)
```

```
frame.pack(pady=10)
```

User Interface(UI)

```
type_var = tk.StringVar(value="Expense")
```

```
category_var = tk.StringVar()
```

```
tk.Label(frame, text="Type").grid(row=0, column=0)
```

```
ttk.Combobox(frame, textvariable=type_var,  
              values=["Income", "Expense"], width=10).grid(row=0, column=1)
```

```
tk.Label(frame, text="Category").grid(row=0, column=2)
```

```
category_entry = tk.Entry(frame, textvariable=category_var)
```

```
category_entry.grid(row=0, column=3)
```

User Interface(UI)

```
tk.Label(frame, text="Amount").grid(row=1, column=0)
```

```
amount_entry = tk.Entry(frame)
```

```
amount_entry.grid(row=1, column=1)
```

User Interface(UI)

```
tk.Label(frame, text="Note").grid(row=1, column=2)
```

```
note_entry = tk.Entry(frame)
```

```
note_entry.grid(row=1, column=3)
```

```
tk.Button(frame, text="Add", command=add_transaction).grid(row=2, column=1, pady=10)
```

```
tk.Button(frame, text="Delete Selected", command=delete_transaction).grid(row=2, column=2)
```


Table

```
columns = ("Date", "Type", "Category", "Amount", "Note")
```

```
tree = ttk.Treeview(root, columns=columns, show="headings")
```

```
for col in columns:
```

```
    tree.heading(col, text=col)
```

```
tree.pack(fill="both", expand=True)
```

Balance

```
balance_label = tk.Label(root, text="Balance: 0.00", font=("Arial", 14))
```

```
balance_label.pack(pady=10)
```

```
load_data()
```

```
root.mainloop()
```